CS4270 Final Synthesis Project Proposal

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Project Title: EquiCity: A 2D City Builder and Resource Allocation Simulator

2. Project Objective and Problem Statement

Modern cities are frequently designed and zoned in ways that disadvantage specific neighborhoods, creating underserved areas with inadequate access to sanitation, timely emergency response, commercial goods, employment, and balanced housing while concentrating environmental burdens like industrial noise pollution near vulnerable residents. EquiCity addresses this problem by simulating potential city layouts on a 2D tile-based map and tracking spatial resource distribution, so users can optimize municipal infrastructure to prioritize areas that need resources most. By creating this program in a game-like manner, we can teach valuable management lessons while keeping users engaged.

The intended end-users are students, educators, and aspiring planners interested in civil engineering, urban development, and city planning. EquiCity improves their workflow and learning experience by providing an interactive 2D WorldBox-style sandbox where they can test zoning configurations, observe immediate feedback on citizen needs, and plan equitable urban layouts that benefit every resident across the city efficiently.

3. Social Issue Alignment

EquiCity explicitly addresses the civic and social issue of Community Resource Allocation and Urban Equity. Rather than treating city building purely as an economic or aesthetic game, the software embeds equitable resource distribution directly into its simulation mechanics. As users place low, medium, and high-density residential housing alongside sewage infrastructure, emergency dispatch stations, commercial shops, and industrial workplaces, EquiCity continuously tracks resource availability and environmental impact for every populated tile. Specifically, the software calculates service capacity, emergency dispatch travel distances, commercial and workplace accessibility, and industrial noise pollution exposure across neighborhoods. By providing visual overlays and diagnostic tools that civil engineers and urban designers need to visualize spatial connections and evaluate the real-world consequences of zoning decisions, EquiCity promotes awareness of systemic urban disparities and empowers users to allocate resources fairly to all residents.

4. Core Features and Sub-Goals

- Procedural 2D Biome Map Generation: Users can generate and initialize a 2D grid map (minimum 128x128 tiles) featuring distinct terrain biomes that shape city layout and infrastructure placement.
- Interactive Zoning and Building Placement: Users can place, inspect, and remove residential housing zones across multiple densities (Low: 1, Medium: 4, High: 10), sewage facilities, emergency dispatch centers, commercial shops, and industrial workplaces.
- Resource and Building Simulation: Users can run simulation updates that map numerical capacity, population demand, emergency dispatch distance, shop and workplace access, sewage load, and noise pollution values directly to buildings and surrounding grid tiles.
- Visual Equity Overlays and JavaFX Metrics: Users can switch between 2D grid visualization modes and view live JavaFX summary metrics highlighting underserved neighborhoods, resource bottlenecks, and overall city equity performance.

- Persistent City State Saving and Loading: Users can save their complete city grid, biome configuration, placed structures, and resource metrics to disk and reload saved states across sessions.
- Sandbox Inspection User Interface: Users can navigate the 2D map interface, select placement tools from a streamlined control panel, and inspect individual tiles and buildings to see detailed citizen resource needs.

5. Planned System Architecture

- Custom Package Name: package equicity; (organized inside the physical equicity subfolder alongside AppDriver.java, package.bluej, and README.md).
- Core Classes and Inheritance: CityGrid manages the 2D coordinate map (minimum 128x128) and spatial queries; Tile represents an individual grid cell with terrain and local resource data; Structure serves as an abstract superclass defining shared coordinates, operational state, and polymorphic resource evaluation methods (such as calculateImpact and getResourceSummary); ResidentialZone, ServiceFacility (further extended by SewageFacility and EmergencyDispatch), and EconomicStructure (further extended by CommercialShop and IndustrialWorkplace) extend Structure to polymorphically model population housing demands, municipal service coverage radii, employment and commercial access, and industrial noise pollution; SimulationEngine coordinates city-wide resource allocation updates and disk serialization.
- Enums and Composition: Custom enums include BiomeType (PLAINS, FOREST, WATER, HILL, MOUNTAIN), HousingDensity (LOW with capacity 1, MEDIUM with capacity 4, HIGH with capacity 10), ResourceCategory (SEWAGE, EMERGENCY_DISPATCH, SHOPS, INDUSTRIAL_WORKPLACES, HOUSING,

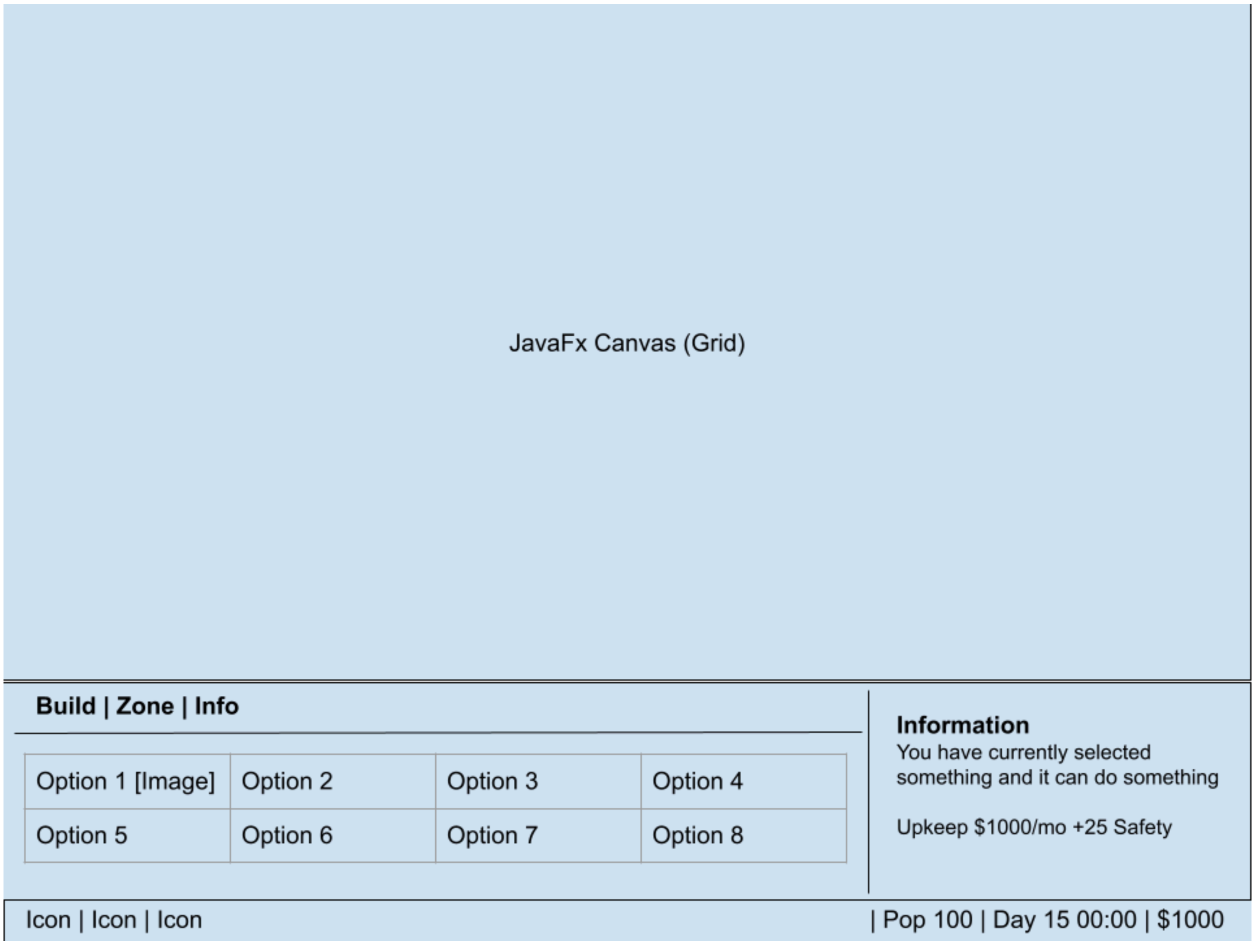
NOISE_POLLUTION), and MapOverlayType (DEFAULT_BIOME, SEWAGE_ACCESS, EMERGENCY_DISTANCE, NOISE_POLLUTION, EQUITY_INDEX). For composition relationships, SimulationEngine HAS-A CityGrid; CityGrid HAS-A 2D array of Tile objects and HAS-A List of Structure instances; each Tile HAS-A BiomeType, HAS-A Map of ResourceCategory values, and optionally HAS-A Structure; ResidentialZone HAS-A HousingDensity.

- GUI and Presentation Layer: The presentation layer is built with JavaFX and strictly separated from the backend equicity domain package using a Model-View-Controller design. AppDriver launches the JavaFX application, while dedicated UI controller and canvas classes render the 2D WorldBox-style tile grid, handle mouse events for placing zones and toggling overlays, and display resource summary panels. Backend classes in package equicity contain no JavaFX imports or UI components, ensuring all simulation calculations and file I/O operate independently of the graphical interface.

6. Team Role Division

- Philip: JavaFX GUI, Event Handling, and Data Visualization (builds AppDriver, the 2D JavaFX map canvas renderer, building placement controls, visual resource overlays, and citizen metrics panels).
- Zachary: Map Generation, Spatial Grid, and State Saving (develops CityGrid, Tile, BiomeType procedural 128x128+ map generation, distance and coverage algorithms, and file I/O persistence for saving and loading city states).
- Braydon: Domain Model Hierarchy and Resource Simulation (designs the abstract Structure superclass, extended building and zone subclasses, polymorphic resource and noise pollution calculations, and equity evaluation logic within package equicity).

UI Example of Main Screen (greyboxed)



JavaFx Canvas (Grid)

Build | Zone | Info

Option 1 [Image]	Option 2	Option 3	Option 4
Option 5	Option 6	Option 7	Option 8

Information
You have currently selected something and it can do something

Upkeep $1000/mo +25 Safety

Icon | Icon | Icon

| Pop 100 | Day 15 00:00 | $1000